

FORT FRANCES, ON, Canada

WMO: 719620

Lat: 48.6511N

Lon: 93.4389W

Elev: 1122

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.4	-23.7	-36.2	0.7	-29.1	-30.1	1.1	-23.5	22.2	13.0	20.2	11.4	3.2	280	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	84.4	68.8	81.4	66.6	78.8	65.1	71.7	80.6	69.7	77.8	67.7	75.5	9.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	110.4	76.7	66.8	102.7	74.4	64.7	95.5	72.5	36.3	81.0	34.5	77.6	32.8	75.7	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	17.9	15.9	-37.9	89.3	6.2	1.6	-42.3	90.4	-46.0	91.4	-49.4	92.3	-53.9	93.4		
			-38.1	75.1	6.2	2.1	-42.5	76.6	-46.1	77.9	-49.6	79.1	-54.0	80.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	4.4	7.4	23.9	37.6	50.8	60.7	65.6	63.4	55.4	42.4	27.1	10.2
	DBStd	24.06	16.08	13.61	14.54	9.56	8.70	6.26	5.58	6.16	8.08	8.45	11.82	13.87
	HDD50	6134	1415	1194	815	383	97	4	0	0	36	267	690	1233
	HDD65	10202	1880	1614	1275	823	448	156	61	107	301	701	1138	1698
	CDD50	1592	0	0	5	10	121	324	485	415	199	32	1	0
	CDD65	185	0	0	0	0	7	26	81	56	14	1	0	0
	CDH74	2039	0	0	3	1	140	310	835	588	148	14	0	0
	CDH80	392	0	0	0	0	25	39	178	124	24	2	0	0
Wind	WSAvg	7.3	7.3	7.3	7.6	8.4	7.8	7.1	6.5	6.3	7.1	8.0	7.7	7.0
Precipitation	PrecAvg	29.3	1.1	0.9	1.3	2.0	3.0	4.5	4.1	3.6	3.5	2.4	1.4	1.3
	PrecMax	38.9	3.4	1.8	3.1	5.0	7.4	8.7	10.6	7.7	7.6	5.4	3.9	2.0
	PrecMin	22.9	0.3	0.2	0.2	0.1	0.4	0.5	1.3	1.2	0.7	0.4	0.2	0.5
	PrecStd	3.7	0.7	0.4	0.7	1.2	1.8	2.1	2.2	1.4	1.8	1.5	0.7	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	42.7	67.0	69.6	83.9	84.2	88.0	87.2	83.2	75.1	60.8	40.6
		MCWB	35.1	38.1	55.0	51.1	62.3	67.7	71.3	71.3	67.3	59.0	50.1	35.6
	2%	DB	34.7	37.3	55.0	64.1	78.2	80.4	84.7	83.7	78.4	67.6	53.5	35.7
		MCWB	32.6	33.2	46.0	48.7	59.2	64.7	70.1	69.6	65.2	55.2	45.7	33.4
	5%	DB	32.1	33.0	48.1	59.2	73.3	77.7	81.5	80.2	74.3	61.8	47.9	33.1
		MCWB	30.4	30.5	40.7	46.0	56.9	63.3	67.4	66.9	63.6	52.8	43.1	31.6
	10%	DB	27.6	28.7	42.3	54.9	68.5	74.7	78.7	77.0	70.4	56.6	43.4	29.7
		MCWB	25.8	26.1	36.7	44.0	55.4	62.3	66.1	65.3	61.9	49.8	39.5	27.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.5	37.7	59.7	55.5	66.3	71.6	75.5	73.6	71.3	60.5	52.2	36.6
		MCDB	38.5	42.6	65.1	63.8	78.0	79.4	84.7	83.0	79.0	69.7	56.8	38.9
	2%	WB	33.0	33.9	46.9	50.1	63.2	67.9	72.0	71.4	67.7	56.8	47.1	33.8
		MCDB	34.7	36.4	54.7	60.5	71.5	76.3	80.3	81.2	74.1	66.6	52.6	35.4
	5%	WB	30.1	30.6	41.1	47.2	60.2	65.9	70.0	69.3	65.2	53.7	43.4	31.7
		MCDB	32.2	33.0	46.8	57.9	69.7	73.9	78.4	77.5	72.7	58.8	48.1	33.3
	10%	WB	25.8	26.2	37.1	44.6	57.2	63.8	67.9	67.2	62.7	50.4	39.2	28.1
		MCDB	27.4	28.4	41.9	54.1	66.7	71.9	76.0	74.7	69.0	56.1	42.9	29.6
Mean Daily Temperature Range	5% DB	MDBR	20.5	23.2	22.2	20.7	23.1	21.3	22.5	23.0	21.0	17.1	14.5	16.5
		MCDBR	17.7	23.5	24.9	30.2	32.9	27.0	26.5	26.7	26.6	27.1	21.4	14.5
		MCWBR	15.5	19.1	16.6	17.7	16.6	14.5	14.2	14.4	16.1	16.5	14.8	12.6
	5% WB	MCDBR	16.9	19.3	22.9	26.3	24.0	21.9	22.2	22.2	22.6	21.6	20.3	14.3
		MCWBR	14.9	15.7	16.0	16.6	13.9	13.4	13.4	13.3	14.8	14.2	14.7	12.8
Clear-Sky Solar Irradiance	taub	0.251	0.268	0.309	0.340	0.356	0.374	0.385	0.373	0.355	0.305	0.284	0.254	
	taud	2.363	2.322	2.361	2.402	2.414	2.385	2.377	2.442	2.462	2.585	2.503	2.402	
	Ebn at Noon	265	284	288	289	287	281	276	275	269	270	251	248	
	Edh at Noon	21	28	32	34	35	36	36	33	29	22	19	18	
All-Sky Solar Radiation	RadAvg	408	789	1184	1519	1671	1766	1862	1575	1158	670	368	289	
	RadStd	54	80	108	133	110	141	109	111	97	92	46	30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page